

Experiment

Creating and Analyzing Disk Images Using dc3dd and Autopsy (Alternative to FTK Imager)

1. Aim

To create a forensic disk image and analyze digital evidence using **dc3dd** and **Autopsy** in **Kali Linux**.

2. Objectives

After completing this experiment, students will be able to:

- Understand the concept of **digital forensics and disk imaging**
- Create a **bit-by-bit forensic copy of a storage device**
- Maintain **data integrity using hashing**
- Analyze disk images to identify **files, deleted files, and system artifacts**

3. Introduction / Theory

Digital Forensics

Digital forensics is the process of **identifying, collecting, preserving, analyzing, and presenting digital evidence** from computers or storage devices.

Digital evidence may include:

- Documents
- Emails
- Browser history
- Logs
- Deleted files

To preserve evidence, investigators create a **forensic disk image**.

Disk Imaging

Disk imaging is the process of creating an **exact replica of a storage device**.

Characteristics of forensic disk images:

- Bit-by-bit copy of storage
- Preserves deleted files
- Maintains data integrity
- Allows investigation without altering original evidence

Hashing in Forensics

Hashing is used to verify the **integrity of digital evidence**.

Common hash algorithms:

Algorithm	Purpose
MD5	evidence integrity
SHA1	verification
SHA256	stronger security

If the hash value of the original disk and the disk image match, the **evidence is considered intact**.

4. Tools Used

Tool	Purpose
Kali Linux	forensic analysis environment
dc3dd	disk image acquisition
Autopsy	forensic investigation and analysis

5. System Requirements

Component	Specification
Processor	Intel i3 or higher
RAM	Minimum 4 GB
Disk	20 GB free space
Software	Kali Linux VM

6. Algorithm

1. Start the Kali Linux environment.
2. Generate activity on the system to create forensic artifacts.
3. Identify the disk partition using system commands.
4. Create a forensic disk image using dc3dd.
5. Generate hash values for integrity verification.
6. Start the Autopsy forensic analysis tool.
7. Import the disk image into Autopsy.
8. Analyze files, logs, and deleted artifacts.
9. Extract relevant digital evidence.

7. Procedure

Step 1: Open Terminal in Kali Linux

Press:

Ctrl + Alt + T

This opens the terminal.

Step 2: Create Sample Evidence

Generate a file that will later be discovered during forensic analysis.

```
echo "Cybersecurity Lab Evidence" > evidence.txt
```

Verify the file creation.

```
Ls
```

Step 3: Identify Disk Partition

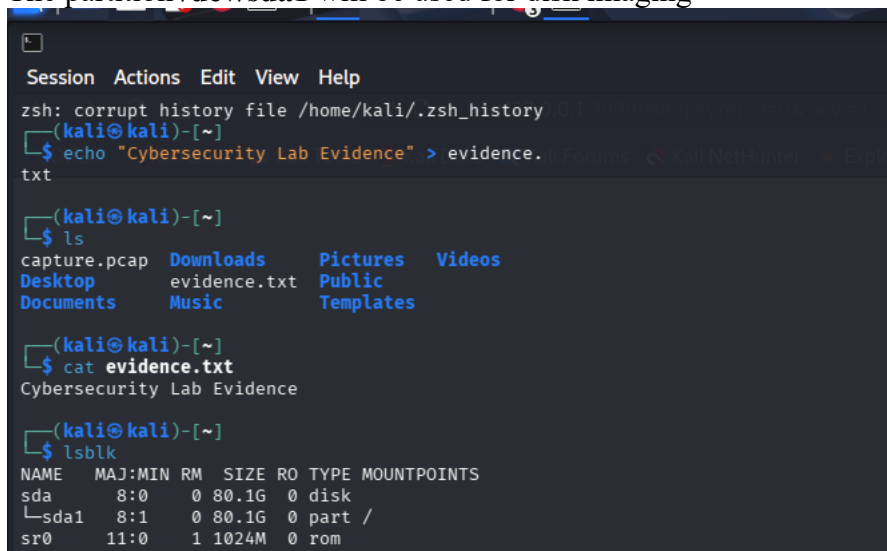
List storage devices.

```
lsblk
```

Example output:

```
sda
├─sda1
└─sda2
```

The partition **/dev/sda1** will be used for disk imaging



```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history (0)
(kali@kali)-[~]
$ echo "Cybersecurity Lab Evidence" > evidence.txt
(kali@kali)-[~]
$ ls
capture.pcap  Downloads  Pictures  Videos
Desktop      evidence.txt  Public
Documents    Music       Templates
(kali@kali)-[~]
$ cat evidence.txt
Cybersecurity Lab Evidence
(kali@kali)-[~]
$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 80.1G 0 disk
├─sda1 8:1    0 80.1G 0 part /
└─sda2 8:1    0 80.1G 0 part /
sr0   11:0    1 1024M 0 rom
```

Install dc3dd

```
(kali㉿kali)-[~]
└─$ sudo apt install dc3dd
Installing:
  dc3dd

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not U
pgrading: 1757
  Download size: 117 kB
  Space needed: 489 kB / 60.2 GB available

Get:1 http://kali.download/kali kali-rolling/main
amd64 dc3dd amd64 7.3.1-4 [117 kB]
Fetched 117 kB in 5s (21.9 kB/s)
Selecting previously unselected package dc3dd.
(Reading database ... 454289 files and directorie
s currently installed.)
Preparing to unpack .../dc3dd_7.3.1-4_amd64.deb .
..
Unpacking dc3dd (7.3.1-4) ...
Setting up dc3dd (7.3.1-4) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.3) ...

(kali㉿kali)-[~]
└─$ dc3dd --help
usage:
_____

        dc3dd [OPTION 1] [OPTION 2] ... [OPTION N
]

                *or*

        dc3dd [HELP OPTION]

        where each OPTION is selected from the ba
sic or advanced
        options listed below, or HELP OPTION is s
elected from the
        help options listed below.
```

Step 4: Create Disk Image

Step 1: Create a Small Disk File

`dd if=/dev/zero of=practice_disk.dd bs=1M count=100`

```
(kali㉿kali)-[~]
└─$ dd if=/dev/zero of=practice_disk.dd bs=1M cou
nt=100
100+0 records in
100+0 records out
104857600 bytes (105 MB, 100 MiB) copied, 0.28316
5 s, 370 MB/s
```

Explanation:

Parameter	Description
if	input disk
of	output image file
hash	generate MD5 hash
log	acquisition log file

Creates a **100 MB disk**.

Step 2: Mount the Disk

```
mkfs.ext4 practice_disk.dd
```

```
(kali㉿kali)-[~]
└─$ mkfs.ext4 practice_disk.dd
mke2fs 1.47.2 (1-Jan-2025)
Discarding device blocks: done

Creating filesystem with 102400 1k blocks and 255
84 inodes
Filesystem UUID: 514f4c47-066d-4a31-8af3-f13b33b7
5b90
Superblock backups stored on blocks:
    8193, 24577, 40961, 57345, 73729

Allocating group tables: done

Writing inode tables: done

Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting inf
ormation: done
```

Step 5: Verify Image Creation

Check the created image.

```
ls -lh /home/kali/disk_image.dd
```

```
(kali㉿kali)-[~]
└─$ ls -lh /home/kali/disk_image.dd
-rw-r--r-- 1 root root 9.8G Mar 15 11:59 /home/ka
li/disk_image.dd
```

Check log file.

```
cat /home/kali/acquisition.log
```

The log will contain:

- image size
- hash value
- acquisition details

```

(kali㉿kali)-[~]
$ cat /home/kali/acquisition.log

dc3dd 7.2.646 started at 2026-03-15 11:52:30 -0400
0
compiled options:
command line dc3dd if=/dev/sda1 of=/home/kali/disk_image.dd hash=md5 log=/home/kali/acquisition.log
device size: 167966702 sectors (probed), 85,998,951,424 bytes
sector size: 512 bytes (probed)
10503356416 bytes ( 9.8 G ) copied ( 12% ), 432.546 s, 23 M/s

input results for device `/dev/sda1':
20514368 sectors in
0 bad sectors replaced by zeros
ce827a581c4dfce5b976603df85bb5e0 (md5)

output results for file `/home/kali/disk_image.dd':
20514368 sectors out

dc3dd aborted at 2026-03-15 11:59:42 -0400

```

Step 6: Start Autopsy

Launch the forensic analysis tool.

Autopsy

```

(kali㉿kali)-[~]
$ autopsy

=====
Autopsy Forensic Browser
http://www.sleuthkit.org/autopsy/
ver 2.24
=====

Evidence Locker: /var/lib/autopsy
Start Time: Sun Mar 15 12:01:18 2026
Remote Host: localhost
Local Port: 9999

Open an HTML browser on the remote host and paste
this URL in it:

http://localhost:9999/autopsy

Keep this process running and use <ctrl-c> to exit

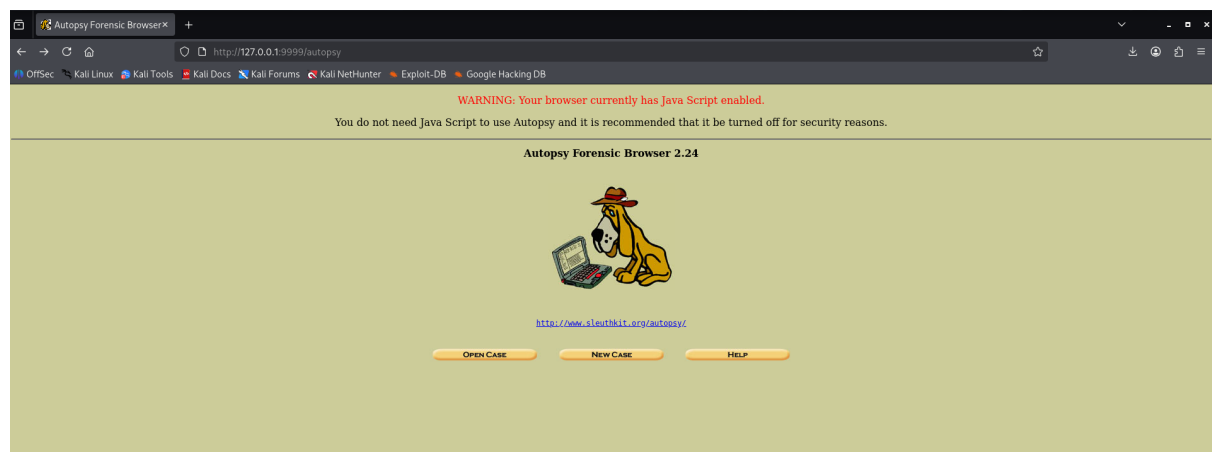
```

```
(kali㉿kali)-[~]  
$ sudo autopsy  
[sudo] password for kali:  
  
Autopsy Forensic Browser  
http://www.sleuthkit.org/autops  
y/  
ver 2.24  
  
Evidence Locker: /var/lib/autopsy  
Start Time: Sun Mar 15 12:16:57 2026  
Remote Host: localhost  
Local Port: 9999  
  
Open an HTML browser on the remote host and paste  
this URL in it:  
  
http://localhost:9999/autopsy  
  
Keep this process running and use <ctrl-c> to exit
```

The terminal will display a URL:

`http://localhost:9999/autopsy`

Open this link in the browser.



Step 7: Create a New Case

In the Autopsy interface:

1. Click **Create New Case**
2. Enter case details:

The screenshot shows a web browser window with the title 'Create A New Case'. The address bar shows the URL 'http://127.0.0.1:9999/autopsy?mod=0&view=1'. The browser's tab bar includes 'OffSec', 'Kali Linux', 'Kali Tools', 'Kali Docs', 'Kali Forums', 'Kali NetHunter', 'Exploit-DB', and 'Google Hacking DB'. The main content area has a light green background and a yellow form titled 'CREATE A NEW CASE'. The form contains three sections: 1. 'Case Name' with a text input field containing 'CyberForensicsLab'; 2. 'Description' with a text input field containing 'Creating a new forensic investi'; 3. 'Investigator Names' with two columns of input fields labeled a. through j., where 'a.' contains 'lavanya'. At the bottom of the form are three buttons: 'New Case', 'CANCEL', and 'HELP'.

Step 8: Add Disk Image

Select:

Add Host → Add Image

Browse and select:

`/home/kali/disk_image.dd`

Autopsy will begin analyzing the disk image.

The screenshot shows a web browser window with the title 'Creating Case: cyberForensicsLab'. The address bar shows the URL 'http://127.0.0.1:9999/autopsy?mod=0&view=2&case=CyberForensicsLab&desc=+Creating+a+new+fo'. The browser's tab bar includes 'OffSec', 'Kali Linux', 'Kali Tools', 'Kali Docs', 'Kali Forums', 'Kali NetHunter', 'Exploit-DB', and 'Google Hacking DB'. The main content area has a light green background. It displays the text 'Case directory (/var/lib/autopsy/CyberForensicsLab/) created' and 'Configuration file (/var/lib/autopsy/CyberForensicsLab/case.aut) created'. Below this, it says 'We must now create a host for this case.' and 'Please select your name from the list: lavanya'. At the bottom is a yellow button labeled 'ADD HOST'.

ADD A NEW HOST

1. **Host Name:** The name of the computer being investigated. It can contain only letters, numbers, and symbols.

KaliLabMachine

2. **Description:** An optional one-line description or note about this computer.

forensic disk image investigatio

3. **Time zone:** An optional timezone value (i.e. EST5EDT). If not given, it defaults to the local setting. A list of time zones can be found in the help files.

4. **Timeskew Adjustment:** An optional value to describe how many seconds this computer's clock was out of sync. For example, if the computer was 10 seconds fast, then enter -10 to compensate.

0

5. **Path of Alert Hash Database:** An optional hash database of known bad files.

6. **Path of Ignore Hash Database:** An optional hash database of known good files.

ADD HOST

CANCEL

HELP

Recommended Values

Host Name

KaliLabMachine

Description

Kali Linux virtual machine used for forensic disk image investigation

Time Zone

Leave Blank

Time Skew Adjustment

Leave Blank

Alert Hash Database

Leave Blank

Ignore Hash Database

Leave Blank

Then click:

Next

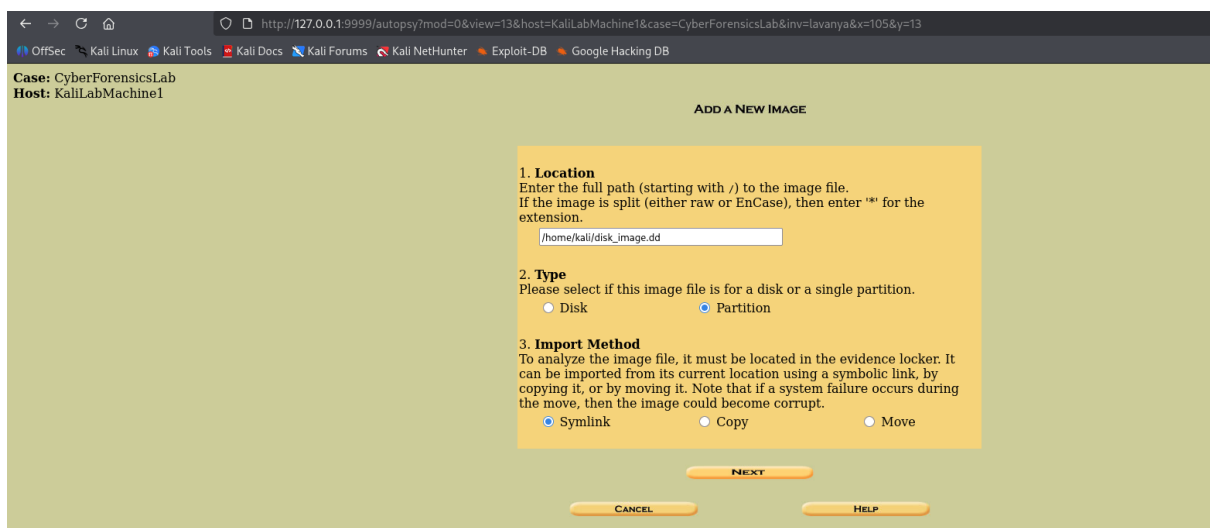
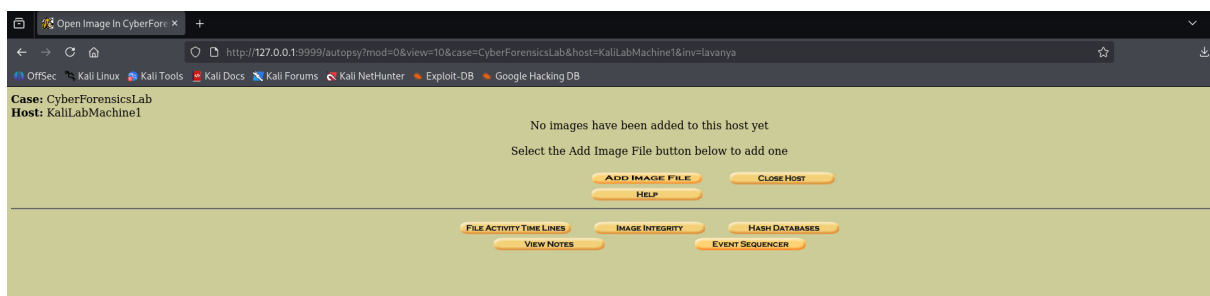
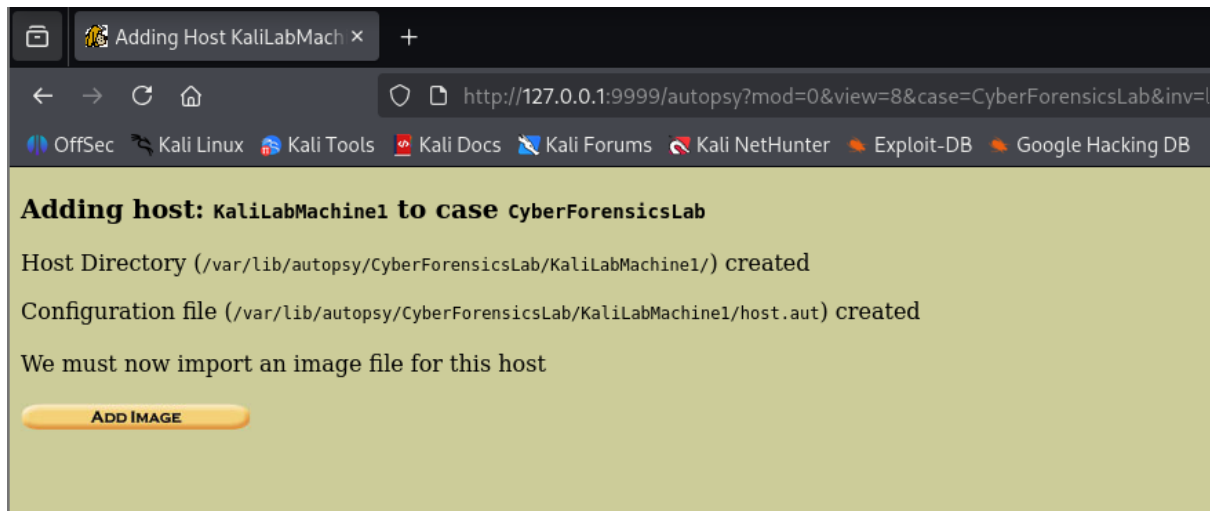


Image File Details

Local Name: images/disk_image.dd

Data Integrity: An MD5 hash can be used to verify the integrity of the image. (With split images, this hash is for the full image file)

- ☐ Ignore the hash value for this image.
- ☒ Calculate the hash value for this image.
- ☐ Add the following MD5 hash value for this image:

☒ Verify hash after importing?

File System Details

Analysis of the image file shows the following partitions:

Partition 1 (Type: ext4)

Mount Point:

File System Type:

ADD

CANCEL

HELP

Calculating MD5 (this could take a while)
Current MD5: ce827a581c4dfce5b976603df85b85e0
Testing partitions
Linking image(s) into evidence locker
Image file added with ID img1
Volume image (0 to 0 - ext - /1/) added with ID vol1

OK

ADD IMAGE

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

Case: CyberForensicsLab
Host: KaliLabMachine1

Select a volume to analyze or add a new image file.

CASE GALLERY			HOST GALLERY	HOST MANAGER
mount	name	fs type		
/1/	disk_image.dd-0-0	ext	details	
ANALYZE			ADD IMAGE FILE	CLOSE HOST
			HELP	
FILE ACTIVITY TIME LINES			IMAGE INTEGRITY	HASH DATABASES
VIEW NOTES			EVENT SEQUENCER	

← → ↺

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

FILE ANALYSIS KEYWORD SEARCH FILE TYPE IMAGE DETAILS META DATA DATA UNIT HELP CLOSE
? X

To start analyzing this volume, choose an analysis mode from the tabs above.

OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB

FILE ANALYSIS
KEYWORD SEARCH
FILE TYPE
IMAGE DETAILS
META DATA
DATA UNIT
HELP
CLOSE

Directory Seek
Enter the name of a directory that you want to view.

View

File Name Search
Enter a Perl regular expression for the file names you want to find.

SEARCH

ALL DELETED FILES

EXPAND DIRECTORIES

Current Directory: [/](#)

Add Note
GENERATE MD5 LIST OF FILES

DEL	Type	NAME	WRITTEN	ACCESSED	CHANGED	Size	UID
	dir / in						
Error Parsing File (Invalid Characters?):							
V/V 5251073: \$OrphanFiles 0000-00-00 00:00:00 (UTC) 0000-00-00 00:00:00 (UTC) 0000-00-00 00:00:00 (UTC) 0000-00-00 00:00:00 (UTC) 0 0 0							
	d / d	del	2025-12-02 22:02:53 (EST)	2026-03-15 01:45:08 (EDT)	2025-12-02 22:02:53 (EST)	4096	0
	d / d	del	2025-12-02 22:02:53 (EST)	2026-03-15 01:45:08 (EDT)	2025-12-02 22:02:53 (EST)	4096	0
	l / l	bin	2025-11-10 04:50:42 (EST)	2026-03-15 01:31:26 (EDT)	2025-12-02 22:02:40 (EST)	7	0
	d / -	boot	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0	0
	d / -	dev	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0	0
	d / -	etc	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0	0
	d / -	home	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0	0

File Browsing Mode

In this mode, you can view file and directory contents.

File contents will be shown in this window.

More file details can be found using the Metadata link at the end of the list (on the right).

You can also sort the files using the column headers

Find in page

☐ Highlight All
☐ Match Case
☐ Match Diacritics
☐ Whole Words

OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB

FILE ANALYSIS
KEYWORD SEARCH
FILE TYPE
IMAGE DETAILS
META DATA
DATA UNIT
HELP
CLOSE

Keyword Search of Allocated and Unallocated Space

Enter the keyword string or expression to search for:

☒ ASCII
☒ Unicode

☐ Case Insensitive
☐ grep Regular Expression

SEARCH

EXTRACT STRINGS
EXTRACT UNALLOCATED

[Regular Expression Cheat Sheet](#)

NOTE: The keyword search runs grep on the image.
A list of what will and what will not be found is available [here](#).

Predefined Searches

SSN2
IP
CC
Date

SSN1

OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB

FILE ANALYSIS
KEYWORD SEARCH
FILE TYPE
IMAGE DETAILS
META DATA
DATA UNIT
HELP
CLOSE

[Sort Files by Type](#)

[View Sorted Files](#)

File Type Sorting

In this mode, Autopsy will examine allocated and unallocated files and sort them into categories and verify the extension.

This allows you to find a file based on its type and find "hidden" files.

WARNING: This can be a time intensive process.

Step 9: Analyze Evidence

